

- 2 (a) The initial total momentum of both balls is not zero.

Since there is no net external force acting on the balls as a system, by the principle of conservation of momentum, the total momentum of both balls must remain unchanged and cannot be zero.

Hence, the balls could not be stationary at the same time.

- (b) By the principle of conservation of momentum,

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$u_A + u_B = v_A + v_B$$

$$4.0 + (-1.0) = v_A + v_B$$

$$v_A + v_B = 3.0 \quad \text{----- (1)}$$

Since collision is elastic,

$$u_B - u_A = v_A - v_B$$

$$(-1.0) - 4.0 = v_A - v_B$$

$$v_A - v_B = -5.0 \quad \text{----- (2)}$$

$$(1) - (2) \quad 2v_B = 3.0 - (-5.0)$$

$$v_B = 4.0 \text{ m s}^{-1} \text{ (shown)}$$

- (c) By Newton's second law, the average force on ball B by ball A is

$$\begin{aligned} F_{\text{net},B} &= \frac{\Delta p_B}{\Delta t} \\ &= \frac{0.50(4.0 - (-1.0))}{0.25} \\ &= 10 \text{ N} \end{aligned}$$

By Newton's third law, the average force on ball A by ball B has the same magnitude of 10 N.

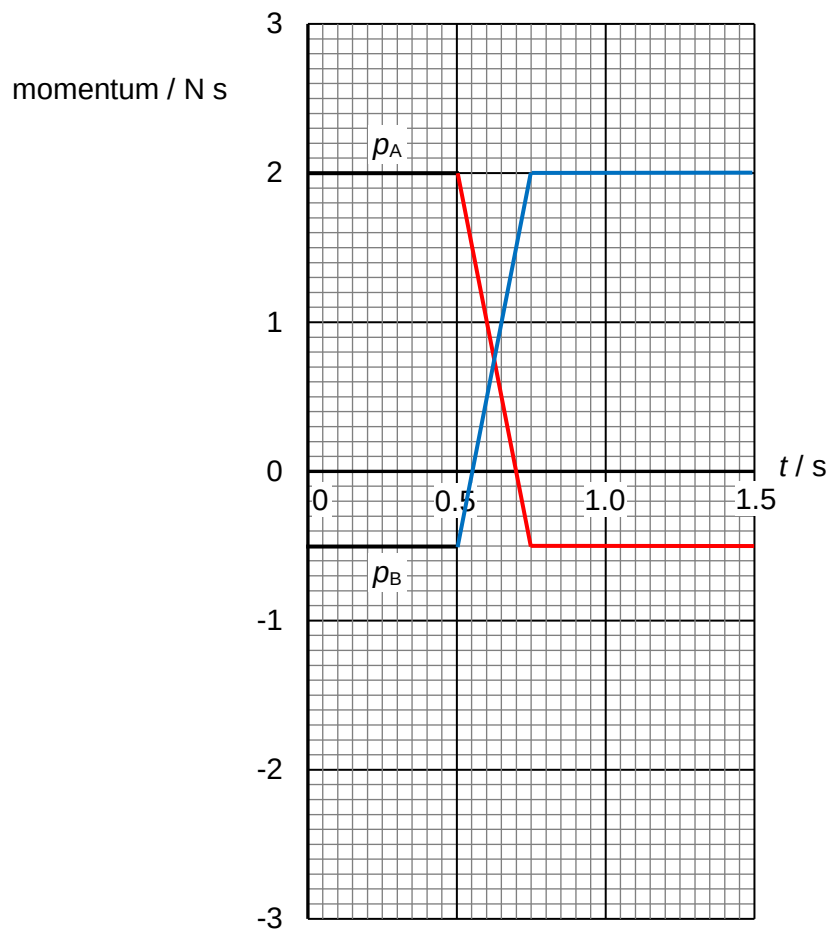
OR

From equation (1) or (2) in part (b),  $v_A = 3.0 - v_B = 3.0 - 4.0 = -1.0 \text{ m s}^{-1}$ .

By Newton's second law, the average force on ball A by ball B is

$$\begin{aligned} F_{\text{net},A} &= \left| \frac{\Delta p_A}{\Delta t} \right| \\ &= \left| \frac{0.50((-1.0) - 4.0)}{0.25} \right| \\ &= 10 \text{ N} \end{aligned}$$

(d)



Balls A and B will exchange velocities and momenta as both balls have the same mass.  
 From (b),  $p_{B,f} = (0.50)(4.0) = 2.0 \text{ N s}$ .

Since duration of collision is 0.25 s, constant final momenta to start from 0.75 s to 1.5 s,  
 with lines joining 0.50 s to 0.75 s during the collision.

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